

Mohak Kapoor

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SUMMARY

ML Engineer with hands-on experience building and deploying production ML systems on edge hardware and cloud infrastructure. Skilled in classical machine learning, deep learning, time-series forecasting, and large language model integration. Engineered production-grade models achieving 96–99% accuracy across intrusion detection, financial forecasting, and optical character recognition. Targeting ML Engineer, Data Science, and AI Engineer roles.

TECHNICAL SKILLS

Languages	Python (Advanced), TypeScript, C++
ML & Modeling	Supervised Learning, Support Vector Machines, LightGBM, XGBoost, CNN, BiLSTM, CRNN, Time-Series Forecasting, Anomaly Detection
LLMs & GenAI	LLM Integration (GPT-4, Gemini), Prompt Engineering, Multimodal Pipelines, MCP, RAG
MLOps & Deployment	FastAPI, REST APIs, Docker, CI/CD (GitHub Actions), MLflow, Cloudflare Workers, Cloudflare Tunnel, Edge Deployment (Raspberry Pi 5)
Frameworks & Data	TensorFlow, PyTorch, scikit-learn, Django, Hugging Face, Pandas, NumPy, Polars, PostgreSQL, SQL
CS Fundamentals	Data Structures & Algorithms, Object-Oriented Programming, Computer Networks

EXPERIENCE

HumanizeIQ

AI Intern

June 2025 – October 2025
Hazlet, New Jersey (Remote)

- Designed and deployed **Model Context Protocol (MCP) servers** on Cloudflare Workers for asynchronous image generation and multimodal document workflows, with persistent object storage via Cloudflare R2.
- Architected recruiter call analysis pipelines integrating **LLMs (GPT-4, Gemini)** for automated transcription analysis and structured report generation — **eliminating 100% of manual post-call processing** and reducing analyst turnaround to zero.
- Engineered **REST APIs** for background job orchestration and asynchronous result retrieval, enabling reliable long-running inference tasks within chat-based LLM workflows.

PROJECTS

METR — Market Exposure Timing Research — Python, Polars, XGBoost, scikit-learn, Feature Engineering 2025

- Engineered a **two-layer meta-labeling framework** on 3,440+ days of multi-asset Indian market data (Nifty 50, GoldBees, USD/INR), using Polars with signal-conditional masking and 30 cross-asset features for trade entry quality classification via XGBoost.
- Validated against **10,000 Monte Carlo simulations** — GoldBees filter achieved **Sharpe 1.48, Calmar 1.87, 66.1% win rate OOS (2024–2025)**, significant at $p < 0.05$ across three independent tests; null results on Nifty and USD/INR consistent with market efficiency theory.

Network IDS Detection Engine — Python, LightGBM, XGBoost, PyTorch, scikit-learn, FastAPI, cuML Sep 2025 – Present

- Engineered a **dual-detection intrusion detection system** (signature-based + unsupervised anomaly detection) on 2.8M rows of CICIDS2017 network traffic, mirroring production platforms like Snort and Darktrace — temporal train/test split to prevent distribution leakage across unseen attack types.
- Supervised: LightGBM **99% accuracy (F1: 0.99)**, XGBoost 97%, PyTorch FFNN 98%, SVM 96.7% — IncrementalPCA **reduced feature space 51%** (69 to 34 features, 98.97% variance retained). Denoising Autoencoder (AUC: 0.7801) outperforms Isolation Forest (AUC: 0.7156) on zero-day threat detection.
- Deployed inference engine as a **FastAPI service on Raspberry Pi 5** (no GPU), exposed publicly via **Cloudflare Tunnel** — modular REST backend with dedicated prediction, routing, and CORS layers.

CaptchaOCR — CAPTCHA Recognition System — Python, PyTorch, CNN, BiLSTM, CTC Loss, Hugging Face 2025

- Developed a CAPTCHA recognition system using a **CNN + BiLSTM (CRNN) architecture with CTC loss**, achieving **96%+ character-level accuracy** on distorted, variable-length sequences via synthetic training corpus generation and noise-augmentation preprocessing.
- Deployed to **Hugging Face Spaces** for public inference, enabling reproducible benchmarking and open community validation beyond offline experimentation.

EDUCATION

Jaypee Institute of Information Technology

B.Tech in Electronics and Communication Engineering

Noida, India
Sept 2022 – July 2026

- Relevant Coursework:** Machine Learning (A+), Deep Learning (A+), Advanced Statistics (A), Computer Systems, Multivariate & Differential Calculus, Control Systems